

NEWINGTON COLLEGE



Preliminary Common Assessment 2001

MATHEMATICS

Extension 1

Time allowed - 2 hours

DIRECTIONS TO CANDIDATES:

- All questions are of equal value.
- All questions may be attempted.
- In every question, show all necessary working.
- Marks may not be awarded for careless or badly arranged work.
- Approved silent calculators may be used.
- The answers to the seven questions in this paper are to be returned in separate bundles clearly marked Question 1, Question 2 etc.
- Each bundle must show the candidate's computer number.
- Start each question on a new page.
- The questions are not necessarily arranged in order of difficulty. Candidates are advised to read the whole paper carefully at the start of the examination.
- Unless otherwise stated candidates should leave their answers in simplest exact form.

OUTCOMES TO BE ASSESSED:

- P2** Provides reasoning to support conclusions which are appropriate in the context;
- P3** Performs routine arithmetic and algebraic manipulation involving surds, simple rational expressions and trigonometric identities;
- P4** Chooses and applies appropriate arithmetic, graphical, trigonometric and geometric techniques;
- P5** Understands the concept of a function and the relationship between a function and its graph;
- PE2** Uses multi-step deductive reasoning in a variety of contexts;
- PE3** Solves problems involving permutations and combinations, inequalities, polynomials, circle geometry and parametric representations.

Question 1 (12 marks)

- (a) Write in simplest form: $2\sqrt{27} - \frac{\sqrt{12}}{2} + 3\sqrt{48}$.
- (b) Solve $|x-3|=11$.
- (c) Factorise fully: $16x^3 - 54y^3$.
- (d) Express as a rational number in simplest terms: $2.03\dot{9}$.
- (e) Give the natural domain for each of the following:
- $f(x) = 3x^2 - 2x + 1$.
 - $g(x) = \sqrt{x^2 - 9}$.
 - $h(x) = \frac{1}{3x-1}$.

Question 2 Start this question on a new page (12 marks)

- (a) Solve for θ , to the nearest minute,
 $\sin \theta = -0.3$, for $0^\circ \leq \theta \leq 360^\circ$.
- (b) Draw a solution for the following inequation on a number line
 $x^2 \leq 16$.
- (c) Express $1 + \frac{2}{\sqrt{3}-1}$, in the form $a + b\sqrt{3}$, where a and b are rational.
- (d) (i) Sketch the curve $y = \frac{1}{2x+1}$, showing all features.
- (ii) If $f(x) = \frac{1}{2x+1}$ is moved horizontally 1 unit to the left, find the equation of the curve, **without** redrawing it. Simplify your answer.

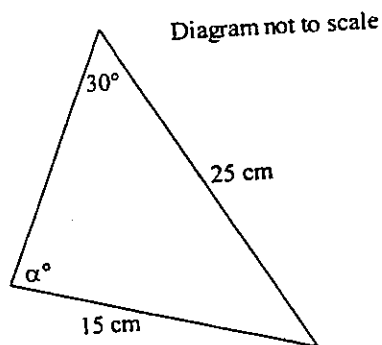
Question 3 Start this question on a new page (12 marks)

- (a) Solve for x , $\tan^2 x + 2\sec x + 1 = 0$, for $0^\circ \leq x \leq 360^\circ$.
- (b) The sides of a triangle are in the ratio 3 : 5 : 6. Find the size of the smallest angle (to the nearest minute).

Q3 cont.... /page 3

Question 3 (cont.)

- (c) Find the value of α , (to the nearest minute).



- (d) Prove $\sin^2 \theta \cos \theta + \cos^3 \theta = \sin(90^\circ - \theta)$.

Question 4 Start this question on a new page (12 marks)

- (a) If $f(x) = x^2 + 2$ and $F(x) = 2x + 3$, then find $f(F(3))$.

- (b) Simplify

(i) $\frac{1}{x^2 - xy} - \frac{1}{x^2 - 2xy + y^2}$

(ii) $\tan \theta + \frac{\cos \theta}{1 + \sin \theta}$

- (c) Find $\sin \theta$, if $\cos \theta = -\frac{12}{13}$ and $90^\circ \leq \theta \leq 180^\circ$.

- (d) Solve for x:

(i) $|2x - 11| = 3x - 4$

(ii) $\frac{2}{x+1} = \frac{x+4}{x+11}$

Question 5 Start this question on a new page (12 marks)

- (a) Solve for x:

$$1 - \sin^2 2x = \frac{1}{2}, \quad -90^\circ \leq x \leq 90^\circ.$$



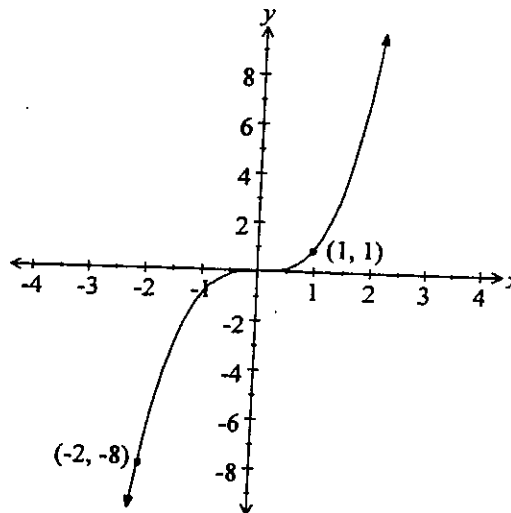
(b)

A rectangular property next to a river has an area of 20 000 m². If a farmer uses the river as one boundary and has only 400 m of fence for the other three sides, what are the dimensions of the field?

Q5...Cont./ page 4

Question 5 (cont.)

- (c) (i) Give the equation of the curve below:
- (ii) Is this function an even or odd function or is it neither? Give reasons.
- (iii) Find the equation for an inverse function.



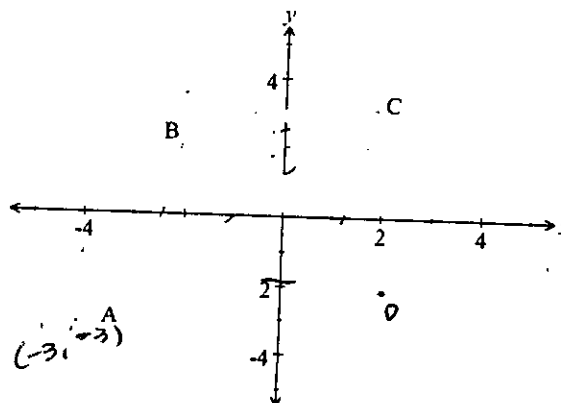
Question 6 Start this question on a new page (12 marks)

- (a) Find the co-ordinates of the point P that divide the interval AB in the ratio $k:1$, where A is $(-3, 5)$ and B is $(-6, -10)$

(i) internally.

(ii) externally.

- (b) In the diagram below, A $(-3, -3)$, B $(-2, 2)$ and C $(2, 3)$ are three points of a parallelogram ABCD. Find the fourth point D.



Q6...cont./page 5

Question 6 (cont.)

- (c) Two ships, A and B, sail from a port simultaneously with A sailing twice as fast as B. Ship A is on a course of 170°T whilst Ship B's course is 290°T . After 1 hour, the two ships are found to be 70 nautical miles apart.

- (i) Draw a diagram to describe the information given above.
- (ii) Find the speed of each ship, in knots, correct to two decimal places and determine also the bearing of B from A at this time, expressed as a true bearing, correct to the nearest degree. (NOTE: a knot is defined as a speed of one nautical mile per hour.).

Question 7 Start this question on a new page (12 marks)

- (a) By completing the square, find the centre and radius of the following circle:

$$x^2 - 3x + y^2 + y - 12 = 0$$

and hence, graph the curve, showing the centre and all intercepts.

- (b) Sketch the region defined by the following inequations:

$$y \leq \sqrt{9 - x^2} \text{ and } y \geq x^2 - 9$$

- (c) Solve for x:

(i) $|3x - 4| + 2|x + 3| \leq 5$

(ii) $\frac{x-1}{x+1} \leq \frac{2x}{2x-1}$

END OF PAPER

Question 1

$$\begin{aligned} \text{a) } 2 \times 3\sqrt{3} - \frac{2\sqrt{3}}{2} + 3 \times 4\sqrt{3} \\ = 6\sqrt{3} - \sqrt{3} + 12\sqrt{3} \\ = 17\sqrt{3} \end{aligned}$$

$$\begin{aligned} \text{b) } |x-3| &= 11 \\ x &= 14 \text{ or } -8 \end{aligned}$$

$$\begin{aligned} \text{c) } 2(8x^3 - 27y^3) \\ = 2(2x-3y)(4x^2+6xy+9y^2) \end{aligned}$$

$$\begin{aligned} \text{d) } 2.03\dot{9} \times 1000 &= 2.039\dot{9} \\ 2.03\dot{9} \times 100 &= 203.\dot{9} \\ 2.03\dot{9} \times 900 &= 1836 \\ \therefore 2.03\dot{9} &= \frac{1836}{900} \\ &= 2\frac{1}{25} \\ &= -\frac{51}{25} \end{aligned}$$

e) i, $x \in \mathbb{R}$

ii, $-3 \leq x \leq 3$??

iii, all $x \in \mathbb{R}$, $x \neq \frac{1}{3}$

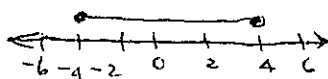
Question 2

a) $\sin \theta = -0.3$

$$\begin{aligned} \frac{1}{x} \quad \text{ref } \theta &= 17^\circ 27' \\ \therefore \theta &= 197^\circ 27' \text{ or } 342^\circ 33' \end{aligned}$$

b) $x^2 - 16 \leq 0$

$$\frac{1}{x} \quad -4 \leq x \leq 4$$



$$\begin{aligned} \text{c) } 1 + \frac{2}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} \\ = 1 + \frac{2\sqrt{3}+2}{3-1} \\ = 1 + \sqrt{3} + 1 \\ = \end{aligned}$$

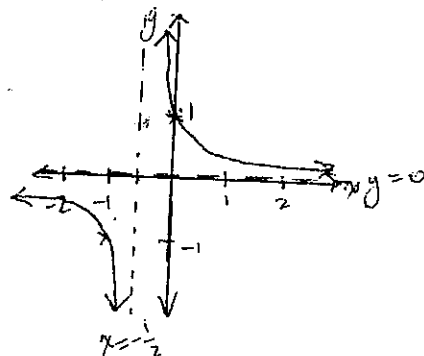
d) i, $D: x \neq -\frac{1}{2}$, $R: y \neq 0$

y -int = 1

x -int = none

check $x = -1$

$y = -1$



$$\begin{aligned} \text{ii, } f(x) &= \frac{1}{2(x+1)+1} \\ &= \frac{1}{2x+2+1} \end{aligned}$$

$$f(x) = \frac{1}{2x+3}$$

Question 3

a) $\tan^2 x + 2\sec x + 1 = 0$

$$\sec^2 x - 1 + 2\sec x + 1 = 0$$

$$\sec^2 x + 2\sec x = 0$$

let $u = \sec x$

$$u^2 + 2u = 0$$

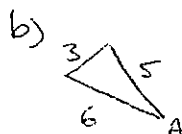
$$u(u+2) = 0$$

$$\therefore u = 0, u = -2$$

$$\therefore \sec x = 0 \text{ or } \sec x = -2$$

No soln

$$\begin{aligned} \cos x &= -\frac{1}{2} \\ \text{ref } x &= 60^\circ \\ x &= 120^\circ, 240^\circ \end{aligned}$$



$$\cos A = \frac{5^2 + 6^2 - 3^2}{2 \times 5 \times 6}$$

$$A = 29^\circ 56'$$

c) $\frac{25}{\sin \alpha} = \frac{15}{\sin 30^\circ}$

$$15 \sin \alpha = 25 \sin 30^\circ$$

$$\sin \alpha = \frac{25 \times \frac{1}{2}}{15}$$

$$\therefore \alpha = 56^\circ 27'$$

Bd)

$$\begin{aligned}
 \text{LHS} &= \sin^2 \theta \cos \theta + \cos^3 \theta \\
 &= \cos^3 \theta \left(\frac{\sin^2 \theta}{\cos^2 \theta} + 1 \right) \\
 &= \cos^3 \theta (\tan^2 \theta + 1) \\
 &= \cos^3 \theta \cdot \sec^2 \theta \\
 &= \cos^3 \theta \cdot \frac{1}{\cos^2 \theta} \\
 &= \cos \theta \\
 &= \sin(90^\circ - \theta) \\
 &= \text{RHS}
 \end{aligned}$$

$$\sin^2 \theta \cos \theta + \cos^3 \theta = \sin(90^\circ - \theta)$$

Question 4

a) $F(3) = 2 \times 3 + 3$
 $= 9$

$f(F(3)) = f(9)$
 $= 9^2 + 2$

$= 83$

b) i) $\frac{1}{x(x-y)} - \frac{1}{(x-y)^2}$

$= \frac{x-y-x}{x(x-y)^2}$

$= \frac{-y}{x(x-y)^2}$

ii) $\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{1 + \sin \theta} \times \frac{1 - \sin \theta}{1 - \sin \theta}$
 $= \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta - \sin \theta \cos \theta}{1 - \sin^2 \theta}$
 $= \frac{\sin \theta \cos \theta}{\cos^2 \theta} + \frac{\cos \theta - \sin \theta \cos \theta}{\cos^2 \theta}$
 $= \frac{\cos \theta}{\cos^2 \theta}$
 $= \frac{1}{\cos \theta}$
 $= \sec \theta$

c) $\cos \theta = -\frac{12}{13}$

$5 \triangle \begin{matrix} 12 \\ 12 \end{matrix} \quad \sqrt{13^2 - 12^2} = 5$
 $\therefore \sin \theta = \frac{5}{13}$

d) i) $|2x-11| = 3x-4$

$\therefore 2x-11 = 3x-4 \quad \text{or} \quad -(2x-11) = 3x-4$

$-7 = x \quad \text{or} \quad 15 = 5x$

$\therefore x = 3, x = -7 \quad x = 3$

ii) $2(x+11) = (x+1)(x+4)$

$2x+22 = x^2+5x+4$

$x^2+3x-18=0$

$(x+6)(x-3)=0$

$\therefore x = -6 \text{ or } 3$

Question 5

a) $1 - \sin^2 2x = \frac{1}{2}$

$-\sin^2 2x = -\frac{1}{2}$

$\sin 2x = \pm \frac{1}{\sqrt{2}} \quad -90^\circ \leq x \leq 90^\circ$

Let $u = 2x \quad -180^\circ \leq u \leq 180^\circ$

$\sin u = \pm \frac{1}{\sqrt{2}}$

$\therefore u = 45^\circ, 135^\circ$

$-45^\circ, -135^\circ$

$\therefore x = \pm 22.5^\circ, \pm 67.5^\circ$



$xy = 20000$

$2x + y = 400$

$\therefore y = 400 - 2x$

$\therefore x(400 - 2x) = 20000$

$2x^2 - 400x + 20000 = 0$

$x^2 - 200x + 10000 = 0$

$(x-100)^2 = 0$

$\therefore x = 100 \text{ m} \quad y = 200 \text{ m}$

c) i) $y = x^3$

ii) odd $f(x) = x^3 \quad f(-x) = (-x)^3$
 $= -x^3$

$= -f(x)$

iii) $x = y^3$

$y = \sqrt[3]{x}$

$\therefore f^{-1}(x) = \sqrt[3]{x}$

QUESTION 6

a) $2 = 3 \quad (-3, 5) \quad (-6, -10)$

$$i) x = \frac{3x - 3 + 2x - 6}{3 + 2}$$

$$= -\frac{21}{5}$$

$$y = \frac{3 \times 5 + 2x - 10}{3 + 2}$$

$$= -1$$

$$\left(-\frac{21}{5}, -1\right)$$

ii) $2 = 3 \quad (-3, 5) \quad (-6, -10)$

$$x = \frac{-3x - 3 + 2x - 6}{-3 + 2}$$

$$= 3$$

$$y = \frac{-3 \times 5 + 2x - 10}{-3 + 2}$$

$$= 35$$

$$(3, 35)$$

i) $\mu_{BC} = \frac{2+2}{2+2}$

$$= \frac{1}{4}$$

$$\therefore \mu_{AD} = \frac{1}{4}$$

Eqn of AD

$$y + 3 = \frac{1}{4}(x + 3) \quad 4y + 12 = x + 3$$

$$x - 4y - 9 = 0 \quad x - 4y - 9 = 0$$

$$AB = \frac{-2 \pm \sqrt{2+3}}{-2+3}$$

$$= 5$$

$$\therefore \mu_{CD} = 5$$

Eqn of CD

$$y - 3 = 5(x - 2) \quad \therefore$$

$$5x - y - 7 = 0$$

Plc of int of AD + CD

$$5x - y - 7 = 0 \quad (1)$$

$$x - 4y - 9 = 0 \quad (2)$$

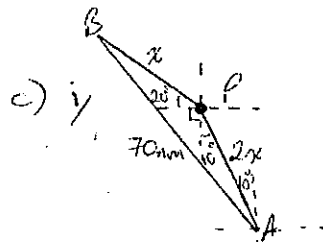
$$(2) \times 5 \quad 5x - 20y - 45 = 0 \quad (2A)$$

$$(1) - (2A) \quad 19y + 35 = 0$$

$$y = -2$$

$$\therefore x = 1$$

$$\therefore D(1, -2)$$



ii) let $x = \text{speed of B}$

$$70^2 = x^2 + (2x)^2 + 2x^2 \cos 120^\circ$$

$$4900 = 3x^2 - 2x^2 \left(-\frac{1}{2}\right)$$

$$4900 = 3x^2 + x^2$$

$$4x^2 = 4900$$

$$x^2 = 1225$$

$$x = 35 \text{ knots}$$

$\therefore$ Ship B $\rightarrow$ 35 knots
Ship A $\rightarrow$ 70 knots

$$\frac{\sin A}{35} = \frac{\sin 120^\circ}{70}$$

$$\sin A = \frac{35 \sin 120^\circ}{70}$$

$$\therefore A = 26^\circ$$

$$26 + 10 = 36^\circ$$

$$360 - 36 = 324^\circ T$$

The bearing of B from A is $324^\circ T$.

QUESTION 7

a) $x^2 - 3x + y^2 + y = 12$

$$x^2 - 3x + \left(\frac{y}{2}\right)^2 + y + \left(\frac{1}{2}\right)^2$$

$$= 12 + \left(\frac{3}{2}\right)^2 + \left(\frac{1}{2}\right)^2$$

$$\therefore \left(x - \frac{3}{2}\right)^2 + \left(y + \frac{1}{2}\right)^2 = \frac{29}{2}$$

$$\therefore \text{centre} \left(\frac{3}{2}, -\frac{1}{2}\right)$$

$$\text{radius } \sqrt{\frac{29}{2}} \text{ or } \frac{\sqrt{58}}{2}$$

$$x\text{-int} \rightarrow x^2 - 3x - 12 = 0$$

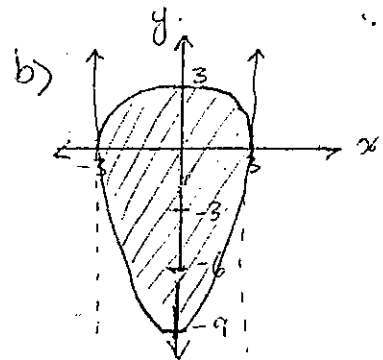
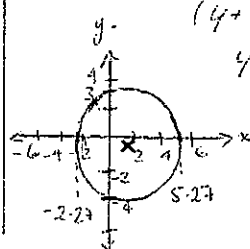
$$x = \frac{3 \pm \sqrt{57}}{2}$$

$$= -2.27 \text{ or } 5.27$$

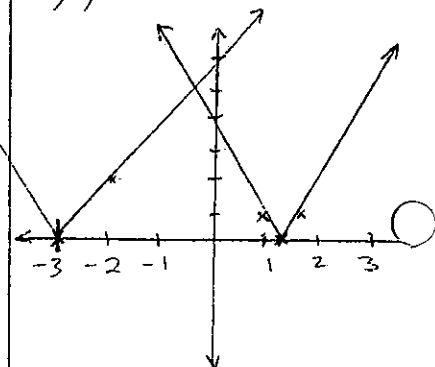
$$y\text{-int} \rightarrow y^2 + y - 12 = 0$$

$$(y + 4)(y - 3) = 0$$

$$y = -4, 3$$



c) i) $|3x - 4| + 2|x + 3| \leq 5$



Case ① $x \leq -3$

$$= (3x - 4) + 2(x + 3) \leq 5$$

$$-3x + 4 - 2x - 6 \leq 5$$

$$-5x - 1 \leq 5$$

$$-5x \leq 4 \quad \therefore$$

$$x \geq -\frac{4}{5}$$

$$\text{Check } x = -\frac{3}{5} \quad \times$$

Case ② $-3 \leq x \leq \frac{4}{3}$

$$3x - 4 - 2(x + 3) \leq 5$$

$$x \leq 15$$

$$\text{check } x = 14 \quad \times$$

Case ③ $x \geq \frac{4}{3}$

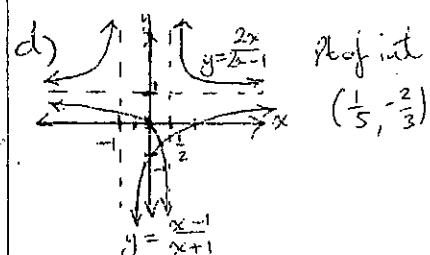
$$3x - 4 + 2(x + 3) \leq 5$$

$$5x + 2 \leq 5$$

$$x \leq \frac{3}{5}$$

$$\text{check } x = \frac{4}{3} \quad \times$$

$\therefore$ No solutions.



$$\therefore -1 \leq x \leq \frac{1}{5}$$